Multiple Sequence Alignment - CodeAlpha Internship

Bioinformatics

The selected sequences were subjected to Multiple Sequence Alignment using Clustal Omega.

The purpose of MSA was to:

- Align multiple insulin protein sequences simultaneously
- Identify conserved and variable regions across species
- Detect evolutionary relationships between sequences

The alignment output showed regions of high conservation, represented by asterisks (*), indicating identical amino acids across all sequences.

This suggests that it performs an essential biological function. The conserved regions likely correspond to functionally important domains necessary for insulin activity.

Minor variations observed in some positions may reflect species-specific adaptations; however, the overall structure and function of the protein remain unchanged.

The high level of conservation confirms that insulin is evolutionarily conserved across species and plays a critical role in glucose metabolism.

The sequence used is as follows:

1. **HUMAN Insulin:**
    MALWMRLLPLLALLALWGPDPAAAFVNQHLCGSHLVEALYLVCGERGFFYTP
    KTRREAED
    LQVGQVELGGGPGAGSLQPLALEGSLQKRGIVEQCCTSICSLYQLENYCN
2. **insulin preproprotein [Pan troglodytes]**
    MALWMRLLPLLVLLALWGPDPASAFVNQHLCGSHLVEALYLVCG
    ERGFFYTPKTRREAEDLQVGQVELGGGPGAGSLQPLALEGSLQKRGIVEQCCT
    SICSLYQLENYCN
3. **insulin [Pan paniscus]**
    MGSETIKPAGTQQPSALQDRLHQKRPSSRSVPRAFASGGLRVPG
    WLDPRPQLCSREDVAGLLKHVGVSPGAPRQGTWPSAGLSPACLPDHCPSAM
    ALWMRLL

PLLALLALWGPDPASAFVNQHLCGSHLVEALYLVCGERGFFYTPKTRREAEDLQ
VGQVELGGGPGAGSLQPLALEGSLQKRGIVEQCCTSICSLYQLENYCN

4. **insulin isoform X1 [Macaca nemestrina]**
MGSETIKPVGAQQPSTLRDRLHQKRPASRSVPRAFASGHCPSAMALWMRLLP
LLALLALWGPDPAPAFVNQHLCGSHLVEALYLVCGERGFFYTPKTRREAEDPQ
VGQVELGGGPGAGSLQPLALEGSLQKRGIVEQCCTSICSLYQLENYCN
5. **insulin isoform X2 [Chlorocebus sabaeus]**
MGSETIKPVGAQQPSTLRDGCIRRGQQAGHCPSAMALWMRLLPLLALLALWG
PDPVPAFVNQHLCGSHLVEALYLVCGERGFFYTPKTRREAEDPQVGQVELGGG
PGAGSLQPLALEGSLQKRGIVEQCCTSICSLYQLENYCN

The results of the MSA are as follows:

clustalo-I20260609-173016-0186-41904140-p1m

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CLUSTAL O(1.2.4) multiple sequence alignment


green_monkey
MGSETIKPVGAQQPSTLRDGCIRRG-----------------------------------   25
primates
MGSETIKPVGAQQPSTLRDRLHQKRPASRSVPRAFASG----------------------   38
placentals
------------------------------------------------------------   0
human_insulin
------------------------------------------------------------   0
primate
MGSETIKPAGTQQPSALQDRLHQKRPSSRSVPRAFASGGLRVPGWLDPRPQLCSREDVAG   60


green_monkey
-------------------------QQAGHCPSAMALWMRLLPLLALLALWGPDPVPAFV   60
primates
-----------------------------HCPSAMALWMRLLPLLALLALWGPDPAPAFV   69
placentals
----------------------------------MALWMRLLPLLVLLALWGPDPASAFV   26
human_insulin
----------------------------------MALWMRLLPLLALLALWGPDPAAAFV   26
primate
LLKHVGVSPGAPRQGTWPSAGLSPACLPDHCPSAMALWMRLLPLLALLALWGPDPASAFV   120

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green_monkey
NQHLCGSHLVEALYLVCGERGFFYTPKTRREAEDPQVGQVELGGGPGAGSLQPLALEGSL   120
primates
NQHLCGSHLVEALYLVCGERGFFYTPKTRREAEDPQVGQVELGGGPGAGSLQPLALEGSL   129
placentals
NQHLCGSHLVEALYLVCGERGFFYTPKTRREAEDLQVGQVELGGGPGAGSLQPLALEGSL   86
human_insulin
NQHLCGSHLVEALYLVCGERGFFYTPKTRREAEDLQVGQVELGGGPGAGSLQPLALEGSL   86
primate
NQHLCGSHLVEALYLVCGERGFFYTPKTRREAEDLQVGQVELGGGPGAGSLQPLALEGSL   180
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green_monkey       QKRGIVEQCCTSICSLYQLENYCN   144
primates           QKRGIVEQCCTSICSLYQLENYCN   153
placentals         QKRGIVEQCCTSICSLYQLENYCN   110
human_insulin      QKRGIVEQCCTSICSLYQLENYCN   110
primate            QKRGIVEQCCTSICSLYQLENYCN   204
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